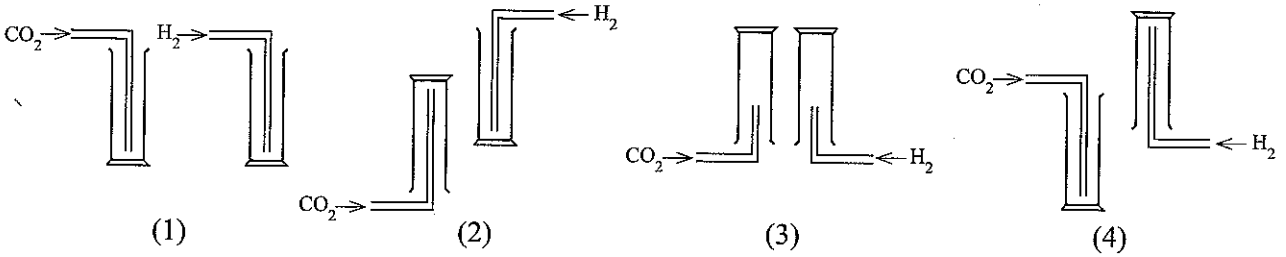
12. Which of the following is the **false** statement about the resistance of a metallic conductor?
(1) It depends on the temperature.
(2) It is directly proportional to the length.
(3) It is inversely proportional to the area of cross section.
(4) It depends on the potential difference across it.

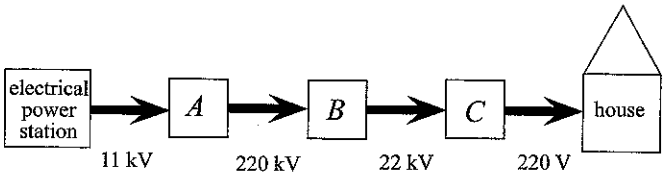
13. Which of the following is the correct way of positioning the delivery tubes to collect two samples of the gases carbon dioxide and hydrogen into gas jars?



(1) (2) (3) (4)

14. Which of the following is **not** a property of synthetic linear polymers?
(1) easily biodegradable (2) combustible
(3) resistant to acids (4) electrical insulators

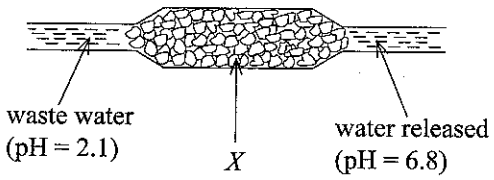
15. This block diagram indicates some main stages of a route of transmitting electricity from an electrical power station to houses. What is the option that gives the transformer types indicated as *A*, *B* and *C* in it respectively?



(1) step-up, step-down, step-down (2) step-up, step-up, step-up
(3) step-down, step-up, step-up (4) step-down, step-down, step-up

16. Which chamber of the human heart has the thickest walls?
(1) left atrium (2) right atrium (3) left ventricle (4) right ventricle

17. The waste water released by a factory was passed through a packing of a solid substance *X* as shown in the figure before emptying into the environment.
Which of the following statement can be true about the waste water and *X*?
(1) Waste water is acidic and *X* is sand.
(2) Waste water is basic and *X* is sand.
(3) Waste water is acidic and *X* is limestone.
(4) Waste water is basic and *X* is limestone.



18. When the cap of a bottle of soda water is opened, carbon dioxide gas bubbles are liberated. More over, carbon dioxide gas bubbles are liberated faster when a bottle of soda water kept outside a refrigerator is opened than when a bottle of soda water kept inside a refrigerator is opened. According to these observations, under which of the following conditions could it be concluded that the solubility of carbon dioxide in water increases?
(1) higher pressure and higher temperature (2) higher pressure and lower temperature
(3) lower pressure and higher temperature (4) lower pressure and lower temperature

19. Which of the following statement is true about the image formed by a plane mirror?
(1) The image is formed on the side of the object itself.
(2) The size of the image depends on the size of the mirror.
(3) The image is laterally inverted.
(4) The image distance depends on the geometrical shape of the mirror.

20. A telecommunication satellite is orbited at a height of 36 000 km from the level of the Earth. What is the time taken in seconds for a radio wave sent by a transmitting station on the Earth to reach the satellite? (Speed of radio waves in air is $3{\cdot}0\times10^{8}$ m s^{-1}.)

(1) $\frac{36000}{3{\cdot}0\times10^{8}}$ (2) $\frac{3{\cdot}0\times10^{8}}{36000}$ (3) $\frac{3{\cdot}0\times10^{8}}{3{\cdot}6\times10^{7}}$ (4) $\frac{3{\cdot}6\times10^{7}}{3{\cdot}0\times10^{8}}$

[*See page three*